IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – [Learnmate AI-personalised learning road map generator]

Domain of Project – [Education/Edtech]

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Recommended font style and size:

Font Style: Arial
Font Size (Heading): 28
Font Size (Content): 20

Problem Statement -

Problem Statement: LearnMate AI – Personalized Learning Roadmap Generator

The Challenge

Students often struggle to identify the right learning path based on their career interests, current skills, educational background, and available study time. With a wide range of online courses, technologies, and learning resources available, choosing what to learn, where to start, and how to progress becomes confusing and time-consuming. Generic learning plans fail to address individual skill levels, learning goals, and time constraints. Without **personalized AI-driven guidance**, students may face difficulties in developing relevant skills, maintaining consistent learning progress, and preparing for their desired careers.

The Objective

Build an **AI-powered Personalized Learning Roadmap Generator** that provides customized learning pathways based on individual student profiles and career aspirations. The solution should:

Personalized Learning Roadmaps – Generate customized learning plans based on students' educational backgrounds, existing skills, career interests, and learning goals.

Skill-Based Recommendations – Identify relevant skills, programming languages, and technologies that students need to learn for their selected career paths.

Time-Based Learning Plans – Recommend structured learning schedules according to students' daily study hours and preferred learning duration.

AI-Powered Guidance – Use IBM Granite AI to generate relevant learning suggestions, topics, and step-by-step career-oriented roadmaps.

Progressive Learning Structure – Organize learning recommendations into structured stages, helping students move from fundamental concepts to advanced topics.

Resource Suggestions – Provide relevant learning topics and resource recommendations to support students throughout their learning journey.

Interactive User Interface – Allow students to enter their details and generate personalized learning roadmaps through a simple and user-friendly Streamlit application.

Proposed Solution -

LearnMate AI is an AI-powered personalized learning roadmap generator designed to help students choose the right learning path according to their interests, current skill level, and preferred learning duration. The application uses a simple Streamlit interface where users enter their learning preferences and requirements. The system processes this information and sends it to IBM watsonx.ai through a deployed IBM Granite model. Based on the user's input, the AI generates a structured learning roadmap containing progressive learning stages, important topics, and recommended learning resources. The generated roadmap is displayed through the Streamlit application in an easy-to-understand format.

The solution includes the following key components:

User Input Collection – Collects the learner's interests, skill level, and preferred learning duration.

Prompt Processing – Converts user information into a structured prompt.

IBM Granite Integration – Uses IBM Granite through IBM watsonx.ai to generate personalized learning content.

Roadmap Generation – Creates a structured learning path with progressive topics and recommended resources.

User-Friendly Interface – Displays the generated roadmap using Streamlit.

LearnMate AI aims to make learning-path planning easier, more personalized, and accessible for students.

Technology Used

Python – core language of the application (Python 3.10 or later).

Streamlit – web interface for the profile form and the roadmap display.

requests – HTTP calls to IBM Cloud IAM and the watsonx.ai deployment API.

python-dotenv – loads WATSONX_API_KEY, WATSONX_URL and WATSONX_DEPLOYMENT_ID from .env.

IBM watsonx.ai – hosts the deployed prompt template used for inference.

IBM Granite – model that generates the personalized roadmap text.

Git & GitHub – version control and project hosting.

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Langflow component Used -

1) Chat Input – Interface where users enter their queries, preferences, or meal details (text/voice/image).

2) IBM watsonx AI Agent - Processes user input using Granite models to generate personalized roadmap for students

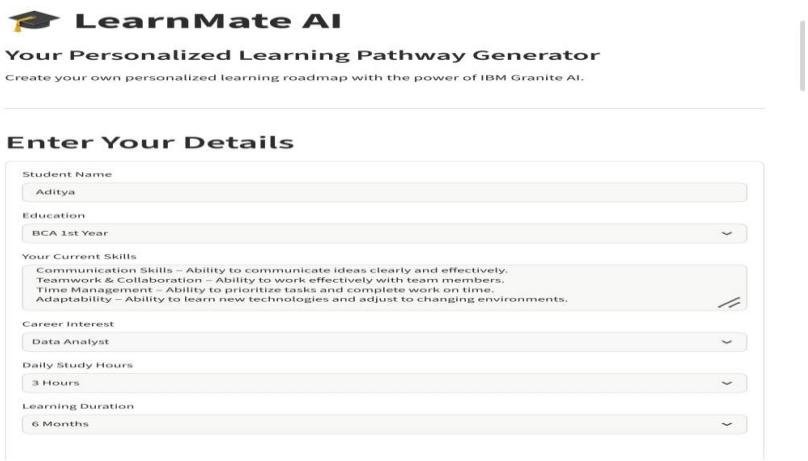
3) Name of IBM Granite model - Granite-4.0-8B-Instruct → Fast, efficient for real-time diet recommendations

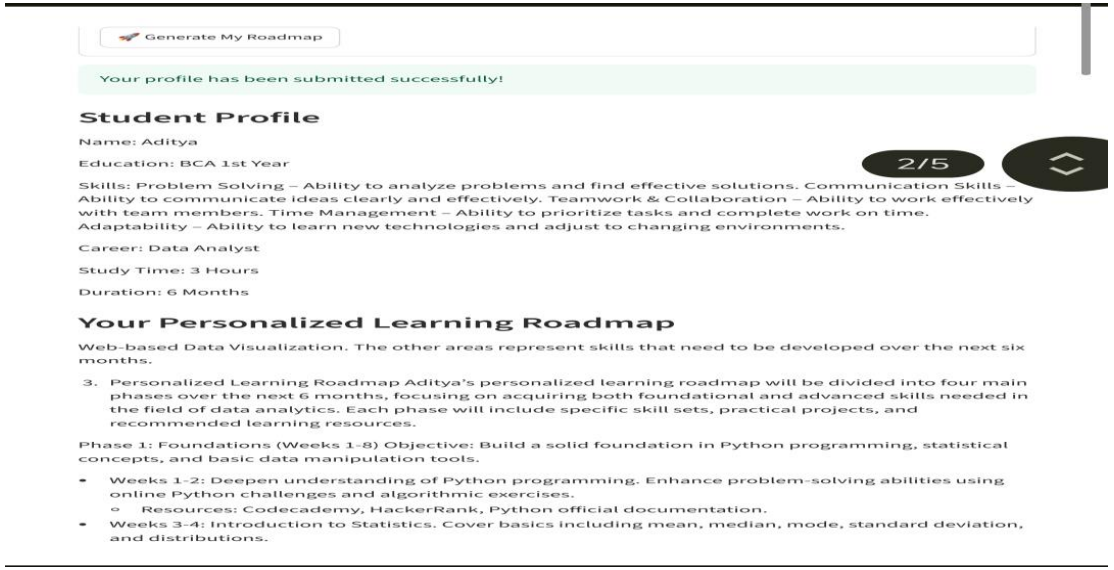
4) Chat output -Displays AI-generated responses such as diet plans, nutritional analysis, and health suggestions.

5) File component - Stores and retrieves user data, meal logs, and nutritional datasets for analysis and tracking.

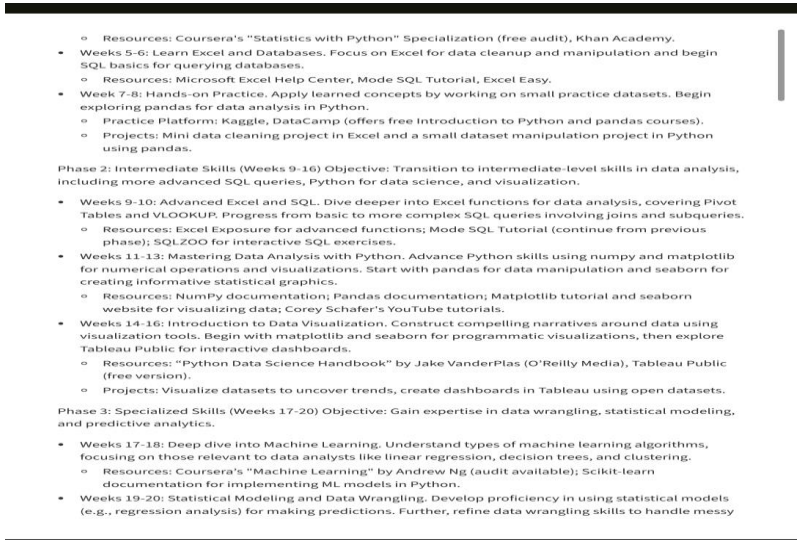
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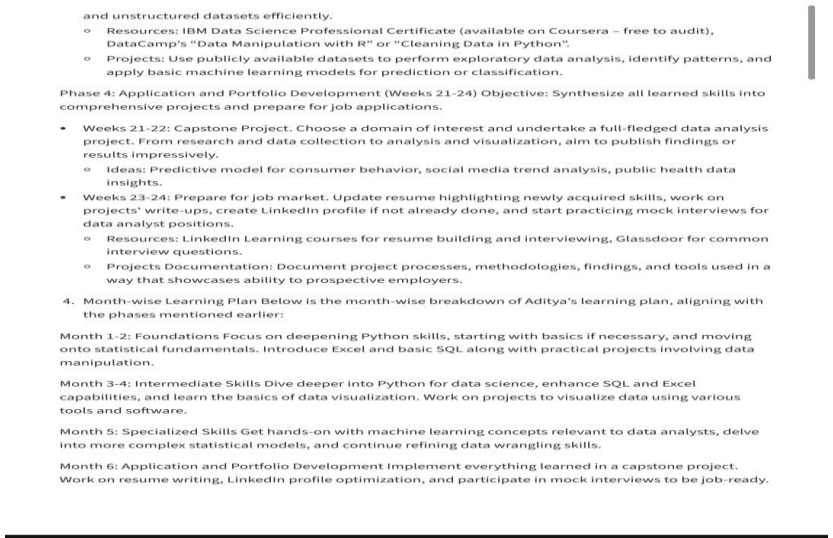
Langflow workflow -



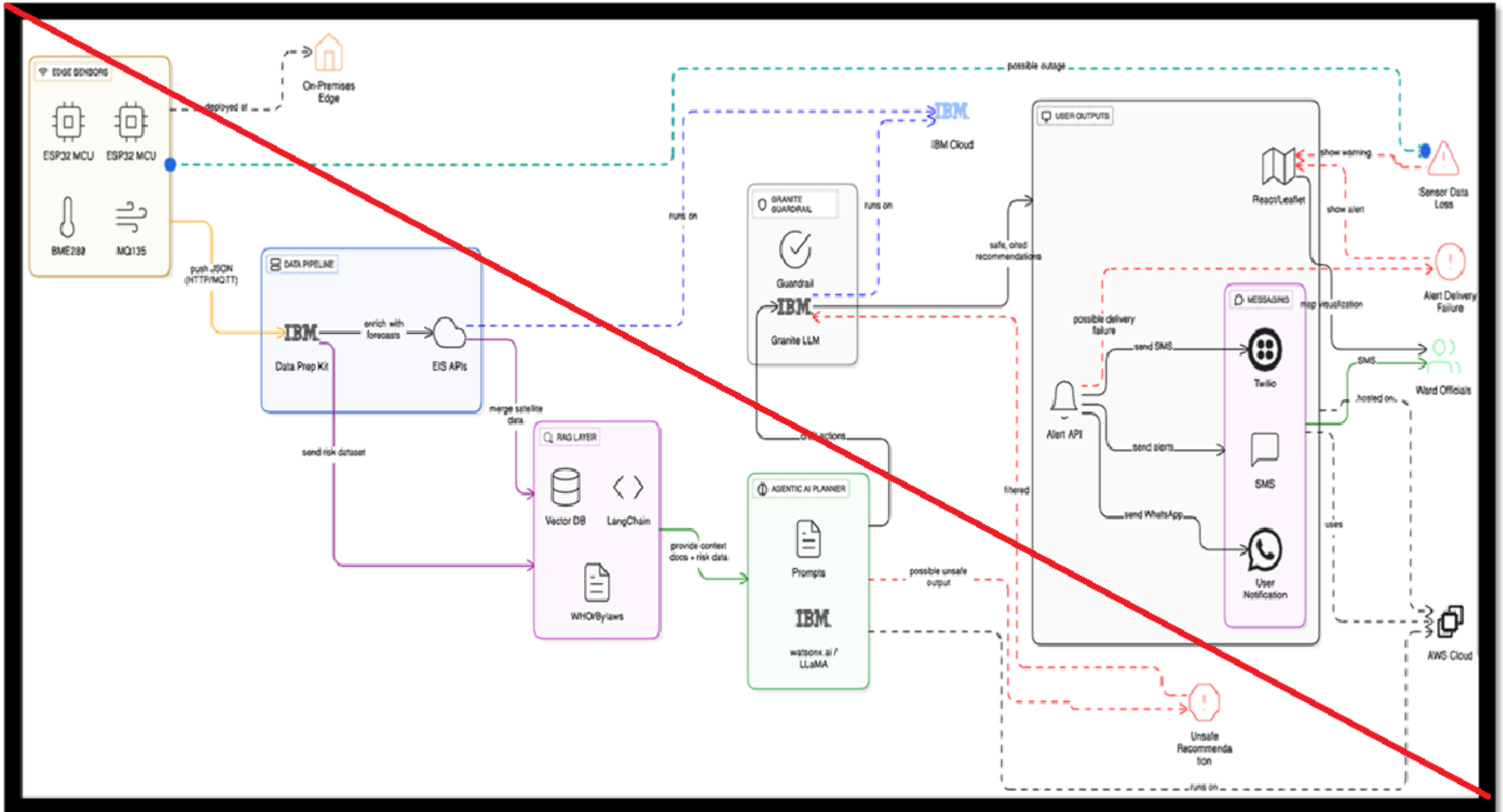


file:///C:/Users/HP/Downloads/mydocument/document/LearnMate%20AI%20%C2%B7%20Streamlit.pdf



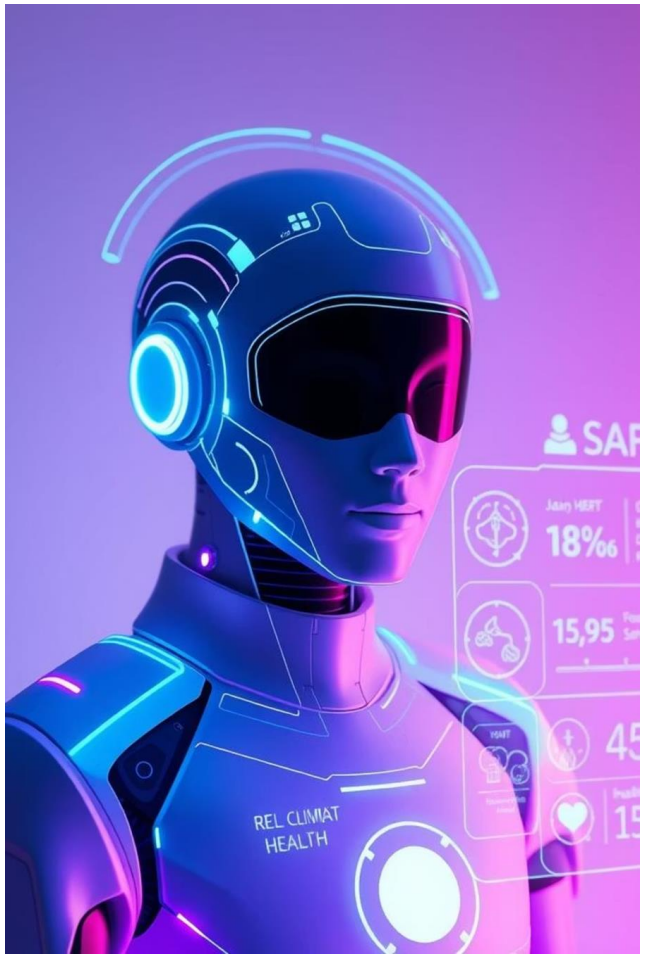


Architecture blueprint



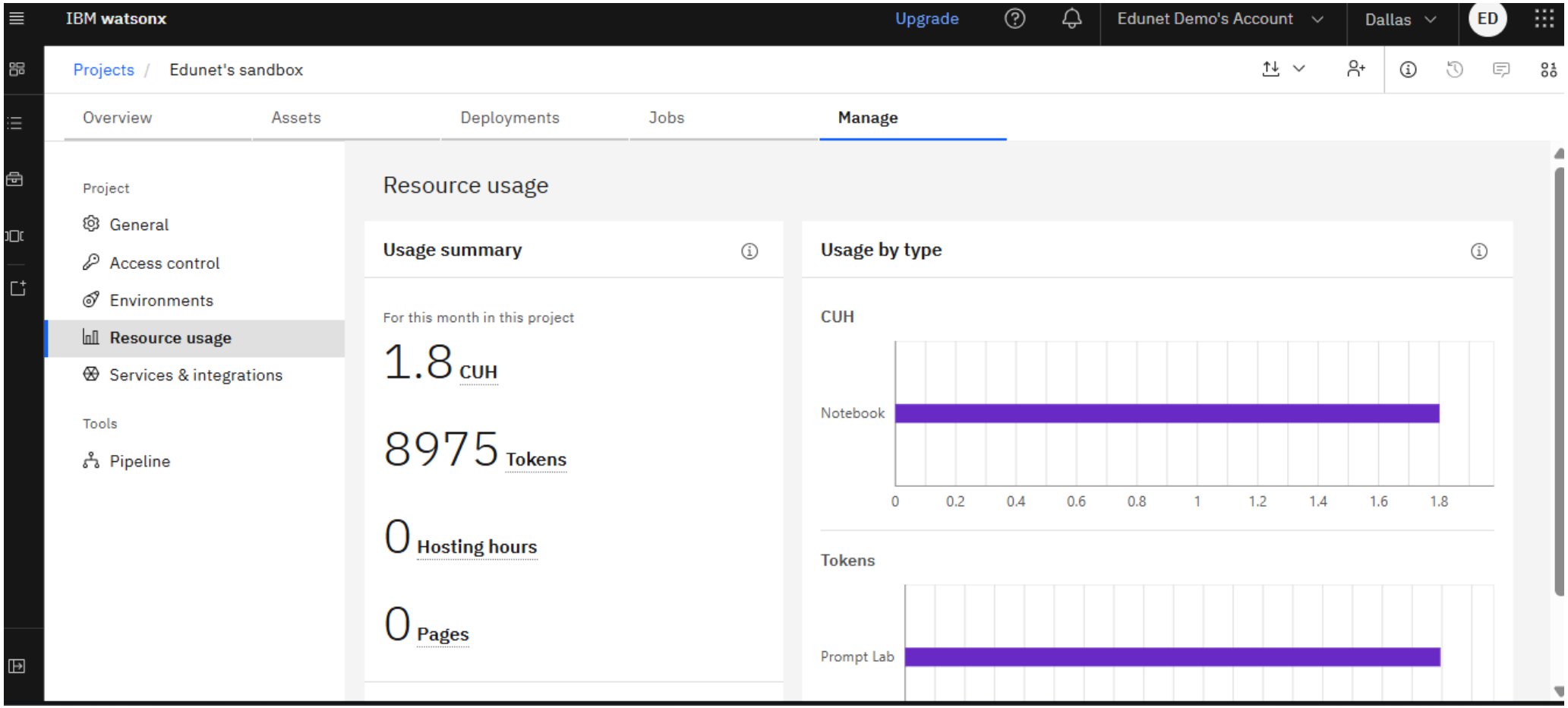
Role of Agentic AI in the solution

Role of Agentic AI in LearnMate AI
LearnMate AI uses Agentic AI concepts to understand user requirements and generate personalized learning roadmaps. The system accepts user input through a Streamlit interface, processes the prompt, and uses the IBM Granite Model through IBM watsonx.ai to generate relevant and structured responses. It helps learners by providing topic-wise recommendations, personalized learning paths, and guidance according to their learning goals. The workflow follows these steps: User Input → Streamlit Interface → Prompt Processing → IBM Granite Model (watsonx.ai) → Personalized Learning Roadmap.
Future Scope: In the future, LearnMate AI can be enhanced with automatic learning progress tracking, continuous roadmap updates, and personalized recommendations based on learner performance.



Token Usage

8975 token are used



Novelty and Uniqueness

The proposed Nutrition Agent stands out by combining **Agentic AI, RAG, and personalization** into a single intelligent system.

Multi-Agent Intelligence – Instead of a single chatbot, multiple specialized agents collaborate to deliver accurate and context-aware nutrition guidance.
RAG-Based Accuracy – Integrates real-time data retrieval with AI generation, reducing misinformation and ensuring trustworthy recommendations.
Hyper-Personalization – Tailors diet plans based on health conditions, cultural preferences, and fitness goals.
Real-Time Feedback Loop – Continuously adapts recommendations based on daily food logs and user behavior.
Multi-Modal Input – Supports text, voice, and image-based meal tracking for better usability.
Preventive Healthcare Focus – Goes beyond diet planning to provide disease-specific nutrition advice.
Scalable & Enterprise-Ready – Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring reliability and governance.

This makes the solution a **smart, adaptive, and proactive digital nutritionist**, not just a static diet recommendation tool.

Git Hub Link

Please create public github repository (Enable Add README button while creating repository) with format –

Please test and Paste the working Github URL –https://github.com/adityakumco-dot/LearnMate-AI?utm_source=chatgpt.com

Make sure you have Uploaded following three files into your github repository

1) app.json file , 2) yourproblemstatement.pdf file 3) your projectpresntation.pptx file 4)any other files

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Future Scope

1.Integration with Wearables & Health Devices –

The system can be integrated with smartwatches and fitness trackers to collect real-time data such as steps, calories burned, heart rate, and sleep patterns. This will enable the Nutrition Agent to provide more accurate and dynamic diet recommendations based on actual user activity and health metrics.

2.AI-Powered Voice Assistant & Multilingual Support –

Enhancing the system with a voice-enabled assistant and support for multiple languages will improve accessibility. Users can interact naturally in their preferred language, making the solution more inclusive and suitable for diverse populations, especially in regions with varied linguistic backgrounds.

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Thank You!

Thank you for your time and interest.